

David Munoz Tord

Geneva, Switzerland
david.munoztord@mailbox.org
david-munoztord.online
munoztd0
david-munoz-tord

Statistical Methods

- Survival Analysis & Cox Models
- Linear & Generalized Mixed-Effects Models
- Hierarchical Bayesian Modeling (brms, Stan)
- Longitudinal & Repeated-Measures Designs
- Multivariate Pattern Analysis
- Reinforcement Learning & Computational Models
- Propensity Score & Causal Inference

Programming

R	10 yrs
Python	5 yrs
SQL	4 yrs
MATLAB	4 yrs
JavaScript	3 yrs

Research Tools

- tidyverse, data.table, DBI
- brms, rstan, JAGS, hBayesDM
- survival, lme4, nlme, emmeans
- ggplot2, plotly, echarts4r
- Quarto, R Markdown, LaTeX

Languages

Profile

Senior R Developer with expertise in clinical trial, software architecture, and production deployment. Specialist in R package development, and CI/CD. Track record of generating CDISC-compliant TFLs and automating regulatory workflows. Strongly involved in open source and technical mentoring.

Experience

Senior R Developer — **02/2025 - Present**
Cytel for Johnson & Johnson

– Leading the R Team at Johnson & Johnson managing the strategic transition from SAS: steering, evangelization, and continuous training of clinical teams
– Intensive development of validated R packages for clinical trial programming, specifically aligned with CDISC standards
– Implementation of automated reporting systems to efficiently generate TFLs (Tables, Figures, Listings) for regulatory submissions
– Design, performance testing, and maintenance of analytical infrastructures (Posit Server) hosting Shiny applications in a strictly regulated environment

Data Science Consultant — **01/2024 - 02/2025**
Freelance - Pharmaceutical & Tech

– Led end-to-end data projects with full autonomy, mapping user requirements to scalable architecture and final deployment
– Developed high-performance custom R Shiny containerized apps, integrating REST APIs and implementing rigorous unit testing: See here for the Chemical Intelligence Platform for Sustainable Design
– Predictive modeling and automation of data pipelines utilizing tidyverse and advanced data visualization libraries (ggplot2, echarts4r, plotly)

Key Contributor & Developer — **2023 - Present**
Opifex

– **echarts4r:** Lead maintainer, high-impact visualization package. Widely adopted across the R community for generating complex, interactive charts (used in production by numerous enterprises and research groups), bridging R data workflows with Apache ECharts.
– **fiRebase:** Lead maintainer, enable secure, scalable backend integrations (authentication, real-time databases) for R Shiny applications, ensuring robust infrastructure for data-driven web apps.

Data Scientist — **03/2023 - 01/2024**
FlowBank

– Led the data team in evolving the architecture and creating the bank's analytical platforms (Data Science, Data Engineering, DevOps)
– Custom Dashboards: Developed containerized R Shiny applications for interactive data exploration, featuring advanced visualizations (plotly, echarts4r) that support rapid decision-making.
– Maintained machine learning pipelines and provided daily technical mentoring to internal cross-functional teams
– Developed internal analytical tools and automated reporting processes for business teams
– Created secure, documented, and tested internal R packages to meet banking sector constraints
– Conducted exploratory analysis of financial data and laid the groundwork for the internal R infrastructure

Lecturer in Biostatistics — **09/2020 - 01/2022**
University of Geneva

– Taught statistics and led practical workshops in R and Python
– Created and maintained course materials and technical documentation for graduate students

French Native

Spanish Native

English Fluent (C2)

German B2

Education

Ph.D. in Computational Neuroscience

03/2019 - 01/2022
University of Geneva — Lemanic Doctoral School
Longitudinal clinical data, hierarchical Bayesian modeling, reinforcement learning. Supervised by Pr. Sander & Pr. Coppin.

MSc in Neuroscience

09/2018 - 03/2020
University of Geneva
Thesis: Denoising of high-resolution fMRI using tICA. Supervised by Pr. Pool.

Complementary Studies in Computer Science

09/2017 - 06/2018
UMass Amherst — USA
Capstone: Machine learning for neuroimaging data.

BSc in Cognitive Science

09/2014 - 06/2017
University of Geneva
Thesis: ERP evidence of auditory spatial attention. Supervised by Pr. Burra.

Recognition

- Global Fellow scholarship (2017)
- Top 2% on StackOverflow (2024)
- Top 12.5% on GitHub (2026)

Research Scientist — 03/2020 - 01/2022
Novo Nordisk

– Clinical Pipelines: Developed large-scale R pipelines for processing and visualizing clinical trial datasets and sensory alteration data.

– Advanced Modeling: Applied Bayesian statistics and machine learning to extract insights from complex biological data.

– Collaboration: Partnered with international teams to define data requirements and deliver analytical outputs.

Open Source Projects and Community

See my GitHub profile for a more comprehensive list.

Author & Lead Developer — Packages
R, Python, Shiny

– Participated in the R Project Sprint 2026 and contributed to base R development alongside **R Core** contributors

– **Pharmaverse member**: Active contributor and maintainer of key NEST framework packages, including **rtables**, **tern**, **junco**, and **formatters**, supporting standardized clinical reporting and analysis workflows.

– **circle (rOpenSci)**: Official maintainer of the CircleCI DevOps integration package for R.

– **R Consortium**: Active member and contributor to initiatives advancing R's ecosystem, interoperability standards, and enterprise adoption.

– **Major contributions**: Contributed source code to reference ecosystem packages, notably **ggplot2**, **brms**, **AFNI**.

Co-President — We Data
Training, Evangelization, Data Science

– **Typr**: Co-creator of a modern typing system for R designed to improve code safety and long-term package maintenance.

– **Open Source Training**: Designed and regularly hosted technical workshops for the community, making all my workshops freely available.

– Regularly organized training sessions, hackathons, and hosted the R community via "**R-Lunches**" at the University of Geneva.

Publications

See my publications page for more. Focus on data-intensive clinical and neuroscience research.

Differential contributions of ventral striatum subregions to the motivational and hedonic components of the affective processing of reward. Pool, E. R., Munoz Tord, D. , Delplanque, S., Stussi, Y., Cereghetti, D., Vuilleumier, P., & Sander, D. *Journal of Neuroscience* (2022) - Paper

Effect of GLP-1 receptor agonist on self-reported liking scales and its neural responses to food stimuli. Coppin, G., Munoz Tord, D., Pool, Cereghetti, D., Golay, A., Sander, D., & Pataky, Z. *Journal of Obesity* (2023) - Paper

Professional References

Gabriel Becker, **Statistical Computing Consultant and R-core contributor**

Joseph Kovach, **Head of Technological Solutions at Johnson & Johnson**

John Coene, **Lead Developer at Opifex**

Pierre Saouter, **Head Data Scientist at FlowBank**